UNIT 10 · ARS/NET PATTERN

MCQ Bank — 70 Questions with Answers

Precision agriculture, variability, mapping, GPS and GNSS (Q1–Q35)

Q1. Precision agriculture means managing a field according to its:

(a) Uniform average
(b) Within-field variability (site-specific)
(c) Colour only
(d) Owner

Q2. The PA management cycle is:

(a) Act → observe
(b) Observe → map → decide → act
(c) Harvest → sow
(d) Buy → sell

Q3. A key benefit of precision agriculture is:

(a) Higher uniform input use
(b) Better input-use efficiency with lower cost and environmental impact
(c) More labour
(d) Larger fields

Q4. Which is NOT an enabling technology of PA?

(a) GPS/GNSS
(b) GIS & remote sensing
(c) Variable-rate technology
(d) Manual broadcasting by hand

Q5. A map produced from a combine sensor plus GPS is a:

(a) Soil pH map
(b) Yield map
(c) Road map
(d) Contour of rainfall

Q6. Grouping areas of similar soil/yield for differential treatment gives:

(a) Management zones
(b) Field boundaries
(c) Grid north
(d) Datum

Q7. Spatial data as grid cells (e.g., imagery) are:

(a) Vector
(b) Raster
(c) Tabular
(d) Analog

Q8. Points, lines and polygons are ___ data:

(a) Raster
(b) Vector
(c) Pixel
(d) Scalar

Q9. GPS was developed by:

(a) India
(b) USA
(c) Russia
(d) China

Q10. The three GPS segments are space, control and:

(a) User
(b) Weather
(c) Fuel
(d) Radio

Q11. The GPS satellites constitute the:

(a) Control segment
(b) Space segment
(c) User segment
(d) Ground segment

Q12. Ground stations that track satellites and update their data form the:

(a) Space segment
(b) Control segment
(c) User segment
(d) Ionosphere

Q13. A GPS receiver on a tractor belongs to the:

(a) Space segment (b) Control segment
(c) User segment (d) Master station

Q14. GPS position is computed by:

(a) Triangulation of angles (b) Trilateration of distances
(c) Guesswork (d) Compass bearing

Q15. The minimum number of satellites for a 3-D position fix is:

(a) 1 (b) 2
(c) 3 (d) 4

Q16. The fourth satellite is needed to correct the:

(a) Orbit (b) Receiver clock error
(c) Weather (d) Antenna

Q17. GPS satellites orbit at an altitude of about:

(a) 400 km (b) 2,000 km
(c) 20,200 km (d) 36,000 km

Q18. Distance from a satellite is found from:

(a) Signal travel time × speed of light (b) Colour
(c) Voltage (d) Temperature

Q19. Satellite geometry quality is expressed by:

(a) DOP (Dilution of Precision) (b) BHN
(c) GF (d) pH

Q20. A LOWER DOP value means:

(a) Worse accuracy (b) Better positional accuracy
(c) No signal (d) More satellites failed

Q21. Which is a GPS error source?

(a) Ionospheric delay (b) Multipath
(c) Clock/orbit error (d) All of these

Q22. Multipath error is caused by:

(a) Signal reflections off objects/ground (b) Battery drain
(c) Wind (d) Rain only

Q23. Autonomous (single-receiver) GPS accuracy is about:

(a) 1 mm (b) 5–10 m
(c) 100 m (d) 1 km

Q24. Differential GPS (DGPS) uses a base station at a ___ point to broadcast corrections:

(a) Random (b) Known (surveyed)
(c) Moving (d) Underground

Q25. DGPS accuracy is roughly:

(a) 50 m (b) Sub-metre to ~3 m
(c) 500 m (d) 1 mm

Q26. Centimetre-level accuracy for auto-steer needs:

(a) Autonomous GPS (b) RTK (carrier-phase)
(c) A compass (d) DOP=10

Q27. RTK accuracy is approximately:

(a) 2–3 cm (b) 2–3 m
(c) 20 m (d) 200 m

Q28. India's satellite-based augmentation system (SBAS) is:

(a) WAAS (b) EGNOS
(c) GAGAN (d) MSAS

Q29. Selective Availability (deliberate GPS degradation) was:

(a) Still on (b) Turned off in 2000
(c) Never used (d) Indian only

Q30. India's regional navigation system is:

(a) GLONASS (b) NavIC / IRNSS
(c) Galileo (d) BeiDou

Q31. NavIC consists of how many satellites?

(a) 3 (b) 7
(c) 24 (d) 31

Q32. NavIC positioning accuracy over India is better than:

(a) 1 mm (b) 10 m
(c) 100 m (d) 1 km

Q33. Which is a GLOBAL GNSS operated by the European Union?

(a) BeiDou (b) Galileo
(c) GLONASS (d) QZSS

Q34. China's global navigation system is:

(a) BeiDou (b) NavIC
(c) GAGAN (d) QZSS

Q35. In farming, GPS is used for all EXCEPT:

(a) Auto-steer & parallel swathing (b) Geo-referencing samples & yield
(c) Variable-rate application (d) Measuring fuel calorific value

---

Answers

Q1 — (b) Within-field variability (site-specific)
Q2 — (b) Observe → map → decide → act
Q3 — (b) Better input-use efficiency with lower cost and environmental impact
Q4 — (d) Manual broadcasting by hand
Q5 — (b) Yield map
Q6 — (a) Management zones
Q7 — (b) Raster
Q8 — (b) Vector
Q9 — (b) USA
Q10 — (a) User — *Three segments = space (satellites), control (ground stations), user (receivers).*
Q11 — (b) Space segment — *Space segment = satellites.*
Q12 — (b) Control segment — *Control segment = ground stations.*
Q13 — (c) User segment — *User segment = receivers.*
Q14 — (b) Trilateration of distances — *Trilateration from distances.*
Q15 — (d) 4 — ≥ *4 satellites — 3 for x, y, z and 1 for the receiver-clock error.*
Q16 — (b) Receiver clock error — *The 4th satellite resolves the receiver clock error.*
Q17 — (c) 20,200 km

Q18 — (a) Signal travel time × speed of light — *Distance = travel time × c.*
Q19 — (a) DOP (Dilution of Precision)
Q20 — (b) Better positional accuracy
Q21 — (d) All of these
Q22 — (a) Signal reflections off objects/ground
Q23 — (b) 5–10 m — *Autonomous ≈ 5–10 m.*
Q24 — (b) Known (surveyed)
Q25 — (b) Sub-metre to ~3 m — *DGPS sub-metre to 3 m.*
Q26 — (b) RTK (carrier-phase) — *RTK carrier phase ≈ 2–3 cm (auto-steer).*
Q27 — (a) 2–3 cm — *RTK ≈ 2–3 cm.*
Q28 — (c) GAGAN — *SBAS = GAGAN in India.*
Q29 — (b) Turned off in 2000
Q30 — (b) NavIC / IRNSS — *NavIC/IRNSS = India.*
Q31 — (b) 7 — *7 satellites in the design constellation (see the 2026 status note in Topic 3b).*
Q32 — (b) 10 m — *< 10 m over India.*
Q33 — (b) Galileo — *Galileo — European Union.*
Q34 — (a) BeiDou — *BeiDou — China.*
Q35 — (d) Measuring fuel calorific value — *Fuel CV is measured by a calorimeter (Unit 9, Topic 6), not GPS — a cross-unit distractor.*

GIS, yield monitors, remote sensing, VRT, ISOBUS and environmental aspects (Q36–Q70)

Q36. A GIS is primarily used to:
(a) Drive the tractor
(b) Capture, store, analyse and display spatial data
(c) Measure temperature
(d) Weld metal

Q37. In GIS, points, lines and polygons are ___ data:
(a) Raster
(b) Vector
(c) Analog
(d) Thermal

Q38. Satellite imagery stored as grid cells is ___ data:
(a) Vector
(b) Raster
(c) Tabular
(d) Scalar

Q39. Building a continuous soil surface from scattered sample points uses:
(a) Buffering
(b) Interpolation (e.g. kriging)
(c) Welding
(d) Sampling only

Q40. The main output of GIS analysis that drives VRT is the:
(a) Yield only
(b) Prescription (application) map
(c) Weather forecast
(d) Fuel bill

Q41. A yield monitor on a combine measures grain flow together with:
(a) Moisture, header width, speed and GPS position
(b) Air pressure
(c) Soil pH
(d) Sound level

Q42. The primary record of within-field variability is the:
(a) Road map
(b) Yield map
(c) Contour map
(d) Cadastral map

Q43. Yield-monitor data must be corrected for:

(a) Flow (time) lag through the combine (b) Colour
(c) Wind chill (d) Fuel type

Q44. Remote sensing acquires crop/soil information:

(a) By soil sampling (b) Without contact, from aerial/satellite platforms
(c) By digging (d) By weighing

Q45. Pixel size / ground sampling distance defines the ___ resolution:

(a) Spectral (b) Spatial
(c) Temporal (d) Radiometric

Q46. The number and width of wavelength bands defines the ___ resolution:

(a) Spatial (b) Spectral
(c) Temporal (d) Radiometric

Q47. How often a satellite images the same area is its ___ resolution:

(a) Spatial (b) Spectral
(c) Temporal (d) Radiometric

Q48. The number of brightness levels (bits/pixel) is the ___ resolution:

(a) Spatial (b) Spectral
(c) Temporal (d) Radiometric

Q49. A passive remote sensor uses:

(a) Its own radar pulse (b) Reflected sunlight
(c) LiDAR (d) Sound

Q50. LiDAR and radar are examples of ___ sensors:

(a) Passive (b) Active
(c) Thermal-only (d) Optical-only

Q51. The platform giving the highest spatial resolution on demand is the:

(a) Satellite (b) High-altitude aircraft
(c) Drone/UAV (d) Balloon

Q52. NDVI is calculated as:

(a) (RED-NIR)/(RED+NIR) (b) (NIR-RED)/(NIR+RED)
(c) NIR×RED (d) NIR/GREEN

Q53. A dense, healthy crop typically has NDVI of about:

(a) -0.5 (b) 0
(c) 0.6–0.9 (d) 5

Q54. NDVI is high for healthy vegetation because leaves:

(a) Absorb NIR (b) Reflect NIR strongly and absorb red
(c) Reflect red (d) Emit X-rays

Q55. Which is another vegetation index?

(a) SAVI (b) BHN
(c) DOP (d) GF

Q56. An above-ground 'N-sensor' on a boom is used for:

(a) Map-based VRT (b) Sensor-based (real-time) VRT
(c) Yield weighing (d) Steering

Q57. Map-based VRT applies inputs using:

(a) A real-time sensor only
(b) A pre-made prescription map + GPS position
(c) Manual judgement
(d) Fixed rate

Q58. Sensor-based VRT differs in that it:

(a) Needs a prior map
(b) Sets the rate in real time from a live sensor, no prior map
(c) Uses no electronics
(d) Is manual

Q59. The VRT control chain is:

(a) Actuator → GPS → sensor
(b) GPS/sensor → controller → actuator → applied rate
(c) Load cell → engine
(d) Map → hand

Q60. The standard tractor–implement communication network is:

(a) ISOBUS (ISO 11783)
(b) ISO 9001
(c) HDMI
(d) RS-232

Q61. Automatic shut-off of boom sections over already-treated areas is:

(a) Section control
(b) Yield mapping
(c) Kriging
(d) Calibration

Q62. Section control mainly reduces:

(a) Fuel octane
(b) Input overlap and waste
(c) Engine noise
(d) Tyre wear

Q63. Variable-rate seeding changes:

(a) Only depth
(b) Seed rate/spacing across the field
(c) Tyre pressure
(d) PTO speed

Q64. The ISOBUS component that performs variable-rate and logging is the:

(a) Universal Terminal only
(b) Task Controller
(c) Alternator
(d) Governor

Q65. Centimetre-accurate guidance for VRT/auto-steer needs:

(a) Autonomous GPS
(b) RTK correction
(c) A compass
(d) No GPS

Q66. A key ENVIRONMENTAL benefit of precision agriculture is:

(a) More chemical use
(b) Reduced nutrient/chemical run-off and waste
(c) More tillage
(d) Higher emissions

Q67. Applying nutrients only where needed reduces:

(a) Yield
(b) Leaching and eutrophication of water bodies
(c) Soil fertility
(d) Rainfall

Q68. Delineating areas of similar soil/yield for differential management creates:

(a) Management zones
(b) Field roads
(c) Contours
(d) Grids of DOP

Q69. A major BARRIER to PA adoption by small farmers is:

(a) Too much rain
(b) High initial cost, data and skill needs
(c) Excess labour
(d) Low sunlight

Q70. Surface analysis in GIS derives, from elevation data:

(a) Slope, aspect and flow
(b) NDVI
(c) Grain moisture
(d) Octane number

Answers

Q36 — (b) Capture, store, analyse and display spatial data
Q37 — (b) Vector
Q38 — (b) Raster
Q39 — (b) Interpolation (e.g. kriging)
Q40 — (b) Prescription (application) map
Q41 — (a) Moisture, header width, speed and GPS position
Q42 — (b) Yield map
Q43 — (a) Flow (time) lag through the combine
Q44 — (b) Without contact, from aerial/satellite platforms
Q45 — (b) Spatial — *Spatial = pixel size.*
Q46 — (b) Spectral — *Spectral = bands.*
Q47 — (c) Temporal — *Temporal = revisit.*
Q48 — (d) Radiometric — *Radiometric = brightness levels.*
Q49 — (b) Reflected sunlight
Q50 — (b) Active
Q51 — (c) Drone/UAV
Q52 — (b) (NIR-RED)/(NIR+RED) — *NDVI = (NIR - RED)/(NIR + RED).*
Q53 — (c) 0.6–0.9 — *Healthy crop 0.6–0.9.*
Q54 — (b) Reflect NIR strongly and absorb red — *Healthy crop reflects NIR and absorbs red.*
Q55 — (a) SAVI
Q56 — (b) Sensor-based (real-time) VRT
Q57 — (b) A pre-made prescription map + GPS position — *Map-based VRT = prescription map + GPS.*
Q58 — (b) Sets the rate in real time from a live sensor, no prior map — *Sensor-based = real-time sensor, no prior map.*
Q59 — (b) GPS/sensor → controller → actuator → applied rate
Q60 — (a) ISOBUS (ISO 11783) — *ISOBUS (ISO 11783).*
Q61 — (a) Section control — *Task Controller section control cuts overlap.*
Q62 — (b) Input overlap and waste
Q63 — (b) Seed rate/spacing across the field
Q64 — (b) Task Controller — *Task Controller does variable rate and logging.*
Q65 — (b) RTK correction — *RTK gives centimetre accuracy.*
Q66 — (b) Reduced nutrient/chemical run-off and waste — *PA lowers run-off, leaching, waste and emissions.*
Q67 — (b) Leaching and eutrophication of water bodies — *Less leaching → less eutrophication.*
Q68 — (a) Management zones
Q69 — (b) High initial cost, data and skill needs
Q70 — (a) Slope, aspect and flow